Twitter or WhatsApp? Public and Private Agent Networks Are Different at the Edge

Anonymous Author(s)
Affiliation
Address
email

Abstract

Public and private agent networks are commonly compared as complete systems, although they differ in several mechanisms at once. We define their typical relationships through how edges are established and the consequences of damaging them, then distinguish three edge dimensions: formation, semantics and attributes. We propose measuring task quality, token cost and exposure separately, and implement configurable proxies in a shared runtime. Existing distributed-information experiments provide a partial illustration. Their reported factorial scores favor open configurations and show a smaller positive contrast for readable stores; a separate sweep finds no stable benefit from relationship descriptions. The decomposition explains how formation and semantics can oppose one another in the canonical public–private comparison. Historical equal-reach controls and reach extensions help locate its scope, but do not establish equivalence or an optimal configuration. Implementation couplings, scoring limitations and incomplete run-level provenance make the numerical evidence preliminary. The framework supports task-specific tests of access and coordination without assuming that a network label, or a null result for one representation of trust, determines which design is best.

1 Introduction

Agent networks can connect strangers through open discovery or coordinate agents acting for established collaborators. Choosing between these public and private designs requires finding useful counterparts, exchanging enough information and controlling interaction cost and scope. Comparing two systems rarely isolates which mechanism explains their difference.

The distinction begins with the relationship. In a typical public network, contact can be initiated unilaterally and a failed exchange carries limited relationship-specific consequences. In a typical private network, contact depends on consent, prior ties or an introduction, and future cooperation can be lost. These conditions often produce different patterns of reach, information sharing and relationship records. Twitter and WhatsApp are intuitive analogies, but visibility alone does not define the distinction.

We separate these patterns into three dimensions of an edge: formation, who can be connected; semantics, what an established connection permits the parties to exchange or inspect; and attributes, what information accompanies the relationship. Knowing more providers and being allowed to inspect a colleague's material are different capabilities. A public–private comparison changes both when it contrasts an open directory with question answering against a bounded team with readable stores. Crossing these settings allows their contributions to be examined separately, while retaining the two original configurations as the motivating comparison.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

This paper makes three contributions. First, it defines public and private relationships through formation and consequences, organized along three edge dimensions (§2). Second, it gives a shared-runtime measurement framework separating quality, token cost and exposure (§3). Third, it maps existing distributed-information evidence to those dimensions and their limits (§4). Configuration choice follows from the mechanisms and task boundaries (§5). The evidence covers one information-pooling family: it neither demonstrates a private advantage on complex work nor establishes an open advantage on every objective. The advance is to clarify what a comparison measures and should isolate next.

2 Public and private networks through three edge dimensions

2.1 Relationship formation and consequences

We use public and private as ideal types of agent relationships. A public relationship can be initiated unilaterally, with limited relationship-specific consequences when an exchange goes badly. A private relationship depends on consent, an existing tie or an introduction, and misconduct can jeopardize future access or cooperation. These definitions concern how a relationship is established and what is at stake in maintaining it. Visibility, coverage, interaction cost and disclosure are consequences to examine, rather than additional defining conditions.

Twitter and WhatsApp provide familiar analogies for these ideal types, not exact models of either platform. Open discovery often supports broad, shallow contact; an established relationship can support repeated interaction and greater access to working context. Neither tendency is a universal property of a platform or a necessary technical constraint. A public directory can contain persistent identities, and a private team can grow.

2.2 Formation, semantics and attributes

Represent a directed interaction edge by $e = (F, S, A)$, where F describes its formation, S its semantics and A its attributes. The decomposition separates three questions that a public–private comparison commonly changes together (Table 1). It is a measurement framework, not a claim that these dimensions exhaust every property of a network or have statistically independent effects.

Table 1: Three dimensions of an edge. The public and private columns describe typical combinations; the last column states the narrower operations tested here.

Dimension	Typical public form	Typical private form	Current operation
Formation	Unilateral contact; broad, shallow reach	Prior ties or introductions; repeated contact	Search and address a directory or a bounded roster
Semantics	Task delegation with returned answers or artifacts	Granted access to working material and persistent context	Answers; Store additionally permits list/read access
Attributes	Platform ratings, certification or outcome records	Shared history, origin, responsibility or tenure	Asserted relationship text and synthetic per-card marks

Formation asks who can be connected. Reach depends on membership, discovery, consent and available paths. The current experiment primarily changes addressability: all directory entries or a roster of twenty. A bounded roster approximates one consequence of private formation. It does not implement the negotiation, endorsement liability or relationship loss in the conceptual definition.

Semantics asks what can cross an established edge. An agent may answer a question using its own context, or allow the requester to inspect stored material. Both can transmit useful evidence. Direct access changes who selects and integrates it, and may reveal context that a question did not elicit. It can also introduce irrelevant material and additional processing cost. Store retains question answering and adds access to short knowledge entries and component artifacts; it does not test unrestricted enterprise-folder access.

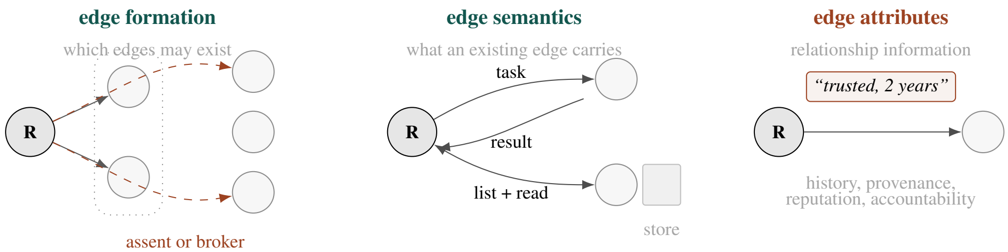

Figure 1: Three questions about an agent-network edge. Formation concerns which connections may form; semantics concerns the exchanges and resources they permit; attributes describe the relationship or its participants. The drawing illustrates the conceptual axes. Our experiment implements roster and access controls, and presents relationship assertions or synthetic marks to the model. It does not implement a full consent, introduction or relationship-loss process. Attributes can inform authorization in other systems; prompt-only presentation is a choice of this instrument, not their definition (Table 1).

Attributes ask what is known about a relationship or counterpart. A stated history, an outcome record and a reputation score convey different kinds of information. Such information can guide partner selection or access decisions in a deployed system. Here, relationship profiles enter as text and reputation marks are assigned synthetically. A null result for these representations does not establish a bound on their general usefulness; it cannot eliminate the attribute dimension or show that learned reputation has no value.

2.3 Typical combinations and diagnostic comparisons

We retain one public configuration, A (open + Sandbox), and one private configuration, D (bounded + Store). The other two cells separate changes that these configurations bundle together. B (open + Store) tests readable access at open reach; C (bounded + Sandbox) tests question answering at bounded reach. B is a diagnostic condition, not a recommendation to open enterprise databases to strangers.

Recorded history, endorsements and stated exit costs are attributes. Using them to create or withdraw contacts changes formation; changing readable resources or retained working context changes semantics. A mechanism can act across dimensions: losing a relationship changes future reach, not merely its description. The runtime's memory and namespace coupling crosses this conceptual boundary. Likewise, exposure can create both distraction and corroboration; the framework leaves their net value to empirical measurement.

3 A controlled measurement framework

3.1 Cross the mechanisms and record what remains coupled

The four-cell design crosses open or bounded formation with Sandbox or Store semantics (Table 2). Within a comparison, agents use the same model client, runtime, coordination kernel, task generator and scoring procedure. The reference implementation uses SharedOS capability grants and application checks to express the conditions in one codebase. Its default model is DeepSeek V4 Flash; the historical $E = 0$ replication used GLM-4.6. Model settings and configuration details are in Appendix B.

Table 2: The factorial conditions. A and D are the fixed public and private combinations; B and C complete the crossed comparison of reach and access interface.

Formation	Sandbox	Store
Open directory	A: typical public	B: diagnostic
Bounded roster	C: diagnostic	D: typical private

The interventions are configurable, but the implementation is not a fully orthogonal realization of the conceptual axes. Open configurations also use per-contact namespaces, no responder memory and a configured delegation depth of zero. Bounded configurations use persistent namespaces, responder memory and a configured depth of five. The requester can keep its own notes in every condition. Formation contrasts therefore estimate this implemented configuration change, rather than a pure effect of social consent or reach alone. Store also changes the artifact-transfer interface. These couplings must remain visible when interpreting the contrasts.

3.2 Tasks, allocation and repeated measurements

Each episode asks a requester to assemble an HTML deliverable from distributed information. The three topics are a conference website, a laboratory dashboard and a travel plan, each with two instances. An instance supplies fourteen required facts and four plausible incorrect claims. Topics vary the content within one information-pooling task family; they do not constitute separate tests of enterprise collaboration, personal errands and social interaction.

The default directory contains 100 cards, of which twenty form the bounded roster. Let H be the number of distinct agents holding required facts. The generator excludes $\mathrm{round}(EH)$ required holders from the roster. Thus E controls holder exclusion, not an exact fraction of missing facts. At $E = 0$, required holders are reachable within the roster, but search effort, memory and access semantics can still differ. The factorial uses $E \in \{0.3, 0.7\}$ and three random seeds. Seeds govern generation and allocation; they are distinct from the relationship-text profiles.

An episode has four stages: T1 is the initial build, T2 revises it, T3 builds a new instance, and T4 revises that instance. These are repeated observations within an episode, not four independent trials. The reference requester has 22 loop iterations for a single-component task; a loop iteration may contain multiple tool calls. This is not a fixed token allowance. Defaults include a search cap of forty cards, no extra edge-cost turns and no relay. Appendix E records the sweep-specific changes.

3.3 Measure quality, cost and exposure separately

Quality. We retain the implementation's name `requirement_f1` for its answer-key score over the deliverable. The checker combines required-value and artifact checks; it is not a general factuality evaluator. Source inspection identified substring matching and HTML-text extraction limitations, so the score must be read as an operational benchmark measure rather than independently validated fact-level precision and recall (Appendix H.2). No reported result uses an LLM judge.

Cost. The client logs prompt and completion tokens for requester and counterpart calls, alongside elapsed time and tool use. A cost comparison must identify its stage, aggregation and included calls. Per-stage and whole-episode totals are different quantities; a bounded–open cost contrast is also different from a Store–Sandbox contrast.

Exposure. Contact counts and resource-access counts are useful proxies for the scope of interaction. Sending a task to an agent can disclose its content; reading a stored entry need not send that agent the full goal. Neither counter alone measures sensitive information leaked or the harm caused. Incorrect-claim exposure and adoption require a separate trace of what was actually visible. The current logging and matching defects prevent the original pollution counts from supporting a fabrication or privacy claim.

3.4 Estimands and evidence status

Writing A–D for mean T1 scores, define the average contrasts

$$\Delta_F = \tfrac{1}{2}[(A - C) + (B - D)], \qquad \Delta_S = \tfrac{1}{2}[(B - A) + (D - C)]. \tag{1}$$

The simple effects and their interaction remain relevant; an average can conceal a formation-dependent semantics effect. Episode-level inference should preserve matched scenario, allocation and seed blocks where they exist.

The numerical summaries below are retained from the existing manuscript. Its run-level outputs are not included in the available repository, and the published analysis scripts do not reproduce every reported pool, delivery filter or paired interval. We therefore report these summaries as

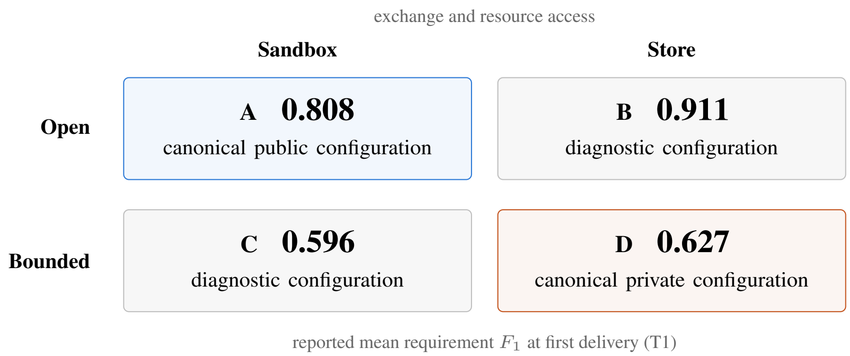

Figure 2: The crossed design separates configuration comparisons from axis contrasts. A and D represent the canonical public and private bundles; B and C provide diagnostic comparisons that vary access semantics within each formation configuration. These are numerical summaries from the existing manuscript, not re-estimated results. Cell-level uncertainty is not reconstructed from rounded means; aggregation and provenance limits are documented in Appendix A.1.

preliminary, not as newly re-estimated results. The reported 582 episodes are an inventory across sixteen sweeps, not the sample size of the main factorial. Appendix A.1 documents the unresolved pooling denominator. Delivery rates, conditional-on-delivery scores and scores assigning zero to non-delivery must be reported separately. A delivery imbalance warning is a diagnostic, not a correction for selection bias.

4 Evidence on distributed information tasks

The existing experiments provide a first use of the framework on distributed-information deliverables. We organize the reported results by the dimension they probe; the full sweep inventory and qualifications are in Appendix A.

4.1 The factorial separates two contributions to the bundle difference

The reported T1 means favor open formation and readable access (Figure 2). The average formation contrast was $\Delta_F = 0.248$ (reported 95% interval $[0.194, 0.302]$), whereas the average semantics contrast was $\Delta_S = 0.066$ ($[0.012, 0.119]$). These reported configuration contrasts are not fixed weights of the framework. Their interval-generating pipeline and relation to corrected task quality remain unverified (Appendices A.1 and H.2).

The canonical public–private difference is $A - D = 0.181$. The diagnostic diagonal $B - C = 0.314$ answers a different question. Algebraically,

$$A - D = \Delta_F - \Delta_S, \qquad B - C = \Delta_F + \Delta_S. \tag{2}$$

For A versus D, the open-formation advantage opposes the readable-access advantage. For B versus C, they reinforce one another. This explains why a bundle difference does not identify either component. The identities hold even when the factors interact. Indeed, subtracting the displayed rounded means gives Store gains of about 0.103 at open reach and 0.031 at bounded reach. These descriptive differences do not establish a statistically reliable interaction without the matched run-level data.

4.2 Coverage is a plausible contributor to the formation contrast

The coverage summary reports open-minus-bounded differences of 0.111 at $E = 0.3$ and 0.279 at $E = 0.7$, with fewer useful sources in the bounded arm as exclusion increases. Its exact arm mapping and filter are unresolved; these are not identified with the factorial formation contrasts (Appendix A.2). The pattern motivates a coverage interpretation without quantifying a uniquely identified reach effect.

The historical $E = 0$ control reported a difference of 0.034 with interval $[-0.067, 0.137]$. That experiment predates the current Sandbox/Store axis, so it is shown separately from the factorial. It

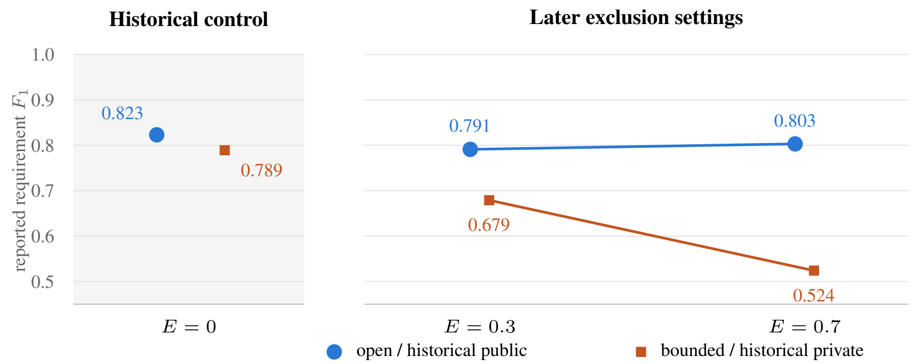

Figure 3: Reported coverage comparisons, with unresolved mappings to factorial cells. E controls the number of required information holders excluded from the roster, not the exact fraction of facts. Points retain the original reported means. The historical $E = 0$ comparison predates the current Sandbox/Store intervention and is shown separately. Reported open-minus-bounded contrasts and 95% intervals are $+0.034$ $[-0.067, 0.137]$ at $E = 0$, $+0.111$ $[0.020, 0.200]$ at $E = 0.3$, and $+0.279$ $[0.210, 0.345]$ at $E = 0.7$. The first interval does not establish equivalence. These summaries do not constitute a single matched experiment. The nonzero rows' exact arm mapping, batch and filter remain unresolved (Appendix A.2); they are not the stratified factorial formation estimates.

did not resolve a score difference; it did not establish equivalence or prove that formation effects vanish at equal reach.

A separate reach extension provides complementary evidence. Allowing an additional access path in bounded configuration C raised its reported score by 0.140 conditional on delivery, and by 0.088 when non-delivery received zero. The implementation permits access beyond the roster; it does not simulate an independently consenting broker or the consequences of an introduction. The result therefore supports extending reach, rather than a claim about social trust (Appendix A.3).

4.3 Readable access raises the reported check score; its cost remains unresolved

Offering Store alongside question answering yielded a higher reported check score than Sandbox alone in both formation configurations. This opposes the original score prediction; it does not establish better information integration or the value of complete folders. The implemented stores contain short entries, and the eight-episode component pilot is too limited to support those claims.

The record does not support a unified cost estimate. The appendix reports a Store increment of 11k tokens with an unresolved stage label. The historical $E = 0$ Flash run reported 49.8k public versus 81.5k private tokens at T3, but these totals are neither the current A/D comparison nor the T1 control. We retain their provenance in Appendix F.2 and do not combine them into an equal-quality cost-dominance claim.

4.4 The tested relationship descriptions did not yield a stable benefit

The separate attribute sweep varied control, trust, accountability, colleague origin, stranger origin and placebo text. The placebo matches origin and stranger in line count and placement, not every profile's token length. No consistent relationship-specific benefit emerged over the placebo. The profile manipulates asserted context, distinct from history earned through successful collaboration. A uniform claim about all counterparts also provides little information for ranking them.

The separate reputation sweep reported scores of 0.821, 0.817 and 0.833 for no marks, informative synthetic marks and random marks. Its contrasts did not resolve a benefit. Because a marked agent could also be the sole holder of a needed true fact, avoiding it could remove useful information. These results limit what the tested representations accomplished; they do not rule out outcome-based reputation with alternative providers (Appendix A.4).

What the evidence supports. The reported factorial scores favor open configurations with a smaller positive Store contrast; the separate text sweep shows no stable benefit. Neither supplies a verified private-network advantage here. The evidence also does not establish that bounded networks are dominated on every objective: the cost basis is unresolved, quality equivalence was not tested, and interaction scope is a separate objective. The original misinformation-adoption and fabrication claims are excluded because their measurement requires correction (Appendix A.5).

5 Task dependence and configuration choice

The decomposition turns a binary label into a question with identifiable parts: does a task benefit from reaching additional holders, inspecting their material, or using relationship information to choose among them? The present tasks mainly reward finding and assembling dispersed facts. Their measured contrasts cannot specify the best configuration for every collaborative setting.

Configuration choice also requires a stated objective. Higher task quality, fewer tokens and a smaller disclosure scope need not select the same cell. For a specified task distribution, the useful output is a quality–cost–exposure comparison at common budgets or quality targets. A configuration is dominated only if an alternative is at least as good on every selected objective and better on one. Failure to detect a quality difference is not evidence of the equivalence needed to make that argument.

Enterprise work motivates a narrower open question: after all necessary holders are reachable, does direct access help integrate evidence that spans documents or versions? For example, a decision may require joining a contract term in one folder to an invoice and an approval record in another. Question answering can in principle transmit the same evidence; whether it omits useful context is a behavior to measure, not a restriction to build into the task. The current short-entry Store does not answer this question.

5.1 Complex-document experiment

Reserved for the new experiment. It has not been conducted.

Methods, results and interpretation will be added after validation.

The framework keeps this question within the semantics dimension. Other settings may place the difficulty in discovery or introductions, but those possibilities need not become additional benchmarks in the present paper. The immediate boundary is whether access semantics matter differently once information integration, rather than missing holders, is the principal difficulty.

6 Scaling, relationship development and limitations

Directory size changes the retrieval problem. Larger directories can add useful providers while making them harder to find. The 50–800-card sweep's accuracy-decline claim was withdrawn because transport-related non-delivery had been counted as zero quality. Corrected means do not establish a scaling law. In the fully delivered subset (three episodes per size), search precision fell from 0.528 to 0.386 while tokens approximately doubled. A controlled rerun must precede million-agent extrapolation (Appendix A.6).

The retained retrieval model illustrates how coverage depends on ranking assumptions (Appendix A.7). Its ρ is an interpolation parameter in a toy rank model, not an empirically estimated correlation or a measured reputation threshold. Outcome feedback could supply information absent from capability descriptions, but whether it improves routing must be tested. Synthetic marks in a task with unique fact holders do not settle that question.

Private networks can grow in membership and in paths. The experiment fixes the roster at twenty; real private networks need not. Their growth may involve new members, additional introductions or broader permissions. Each can change coverage and the costs of maintaining reliable cooperation. The current extra-access-path experiment measures none of the consent, responsibility or declining willingness that a long chain of introductions might involve. It therefore provides no measured private scaling curve to compare with directory growth.

Repeated interaction is present; relationship development remains open. T2 and T4 revisit a deliverable, and T3 reuses an episode's interaction context where the configuration allows it. The study is not a collection of entirely one-shot encounters. It nevertheless does not measure long-term partner selection, learned reputations, relationship formation or relationship loss across episodes. A transition from open search to repeated cooperation is a possible use of the three-axis framework, rather than a process observed here.

The framework is broader than the present instrument. The principal empirical limits are one task family, one main model, bundled configuration changes, imperfect scoring and incomplete run-level provenance. They restrict both causal attribution and generalization. The historical second-model control cannot establish model independence of the main factorial. These limits leave the conceptual decomposition useful while making its numerical contrasts provisional. Confidentiality, adversarial incentives and recourse require dedicated outcomes; they cannot be inferred from contact counts or asserted relationship text.

7 Related work

Network formation and collaboration. Strategic network formation links the creation of ties to their benefits and costs [Jackson and Wolinsky, 1996]; weak-tie theory explains why distant contacts can supply information unavailable within close groups [Granovetter, 1973]. Exploration and exploitation [March, 1991] and transactive memory [Wegner, 1987] provide related accounts of search and knowledge coordination. These theories motivate mechanisms to distinguish. They do not imply that a language model acquires the benefits of a human relationship when prompted to describe one.

Agent interfaces and authority. The Agent2Agent protocol supports task and artifact exchange while allowing agents to keep internal implementation details private [Surapaneni et al., 2025]. Object-capability systems separate authority by explicit references and grants [Miller, 2006]. Our framework uses this distinction between interaction and resource authority to vary what an edge permits. The contribution is the organization of comparisons across network configurations, not a new communication protocol or coordination kernel.

Measuring distributed information use. The hidden-profile paradigm studies groups whose members hold different information [Stasser and Titus, 1985]. Our tasks adapt that motivation to deliverable construction with specified values. Deterministic checks avoid dependence on an LLM evaluator and its possible self-preference [Panickssery et al., 2024], but they introduce their own validity requirements. The scoring and exposure limitations documented here remain relevant even without a learned judge.

8 Conclusion

Public and private agent networks combine differences in relationship formation, information exchange and relationship information. Separating these dimensions makes a system comparison interpretable: a bundle difference need not equal the effect of any one component.

The reported factorial check scores favor open configurations with a smaller positive Store contrast; a separate text sweep shows no stable gain. These summaries provide no verified private-network advantage on the current task family, while their implementation and provenance limits prevent stronger causal or universal conclusions. The next useful comparison concerns access semantics when all necessary holders are reachable and the work requires integrating documents. That experiment remains unrun; its outcome should determine the boundary of the claim rather than be assumed by the framework.

References

Mark S. Granovetter. The strength of weak ties. *American Journal of Sociology*, 78(6):1360–1380, 1973. doi: 10.1086/225469.

Matthew O. Jackson and Asher Wolinsky. A strategic model of social and economic networks. *Journal of Economic Theory*, 71(1):44–74, 1996. doi: 10.1006/jeth.1996.0108.

James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1): 71–87, 1991. doi: 10.1287/orsc.2.1.71.

Mark S. Miller. *Robust Composition: Towards a Unified Approach to Access Control and Concurrency Control*. PhD thesis, Johns Hopkins University, Baltimore, MD, 2006.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pages 68772–68802, 2024. arXiv:2404.13076.

Garold Stasser and William Titus. Pooling of unshared information in group decision making: Biased information sampling during discussion. *Journal of Personality and Social Psychology*, 48(6): 1467–1478, 1985. doi: 10.1037/0022-3514.48.6.1467.

Rao Surapaneni, Miku Jha, Michael Vakoc, and Todd Segal. Announcing the Agent2Agent protocol (A2A). Google Developers Blog, 2025. https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interoperability/. Accessed 2026-09-05.

Daniel M. Wegner. Transactive memory: A contemporary analysis of the group mind. In Brian Mullen and George R. Goethals, editors, *Theories of Group Behavior*, pages 185–208. Springer, New York, 1987. doi: 10.1007/978-1-4612-4634-3_9.

Table 3: Reported experiment ledger. Eleven named sweeps and five smoke sweeps sum to 16 sweeps and 582 episodes. The smoke sweeps are recorded only in aggregate in the available manuscript; their individual names and counts have not been recovered.

Sweep	Varied	Episodes	Role in this revision
`realistic`	2×2 configurations, E	72	Reported main factorial aggregate
`axis2x2`	2×2 configurations, E	72	Second reported factorial batch
`matched-E0`	Historical three-arm design, $E = 0$	27	Coverage-matched historical control
`main`	Historical three-arm design, E	54	Early comparison and read-all diagnostic
`matched-E0-glm`	Second model, $E = 0$	18	Historical model-specific control
`scaling`	Edge cost, relay depth	112	Extra-access and friction comparisons
`seedaxis`	Six text profiles	72	Asserted relationship context
`repscan`	Reputation off/real/random	48	Synthetic reputation marks
`capscan`	Search cap 5/20/80	24	Returned-card budget comparison
`nscan`	Directory size 50–800	60	Original scaling conclusion withdrawn
`k-pilot`	Deliverable components	8	Exploratory composition pilot
five smoke sweeps	Mechanism checks	15	Implementation checks, not effect estimates

A Experiment ledger and supplementary comparisons

The study record reports 16 sweeps, 582 episodes and 2,328 beats. Table 3 retains this accounting across the factorial experiments, historical controls, exploratory sweeps and smoke tests. These totals are not the sample size of the main factorial comparison. An episode contains four successive beats; these are repeated measurements, not four independent samples.

A.1 Evidence provenance and the main aggregate

The numerical results retained here are *reported estimates from the existing study record*. The repositories contain the experiment generator, runtime and analysis code, but not the episode-level outputs needed to reproduce these estimates and intervals. Consequently, a code inspection can establish what a configuration implements without independently certifying its reported empirical effect.

The latest aggregation description assigns the main four-cell means to delivered T1 outputs pooled from `realistic` and `axis2x2`: 144 attempted episodes and 35–36 delivered outputs per cell. An older results paragraph instead described 72 factorial episodes plus 72 attribute episodes. We use the former as the *reported* main-table provenance and keep `seedaxis` separate. The per-run manifests, exact cell counts, profile settings and common aggregation script are still needed to resolve this discrepancy. In particular, the code permits `realistic` to change both relational text and distractor placement; the two batches cannot be certified as identical replications from their names alone.

The reported intervals are percentile bootstrap intervals over episodes. The available factorial script resamples cells separately and reads one `all.json`; it does not implement the documented two-batch, delivered-only aggregate. The exact resampling and filtering procedure behind the retained intervals is therefore unresolved. Both adding a second batch and changing the treatment of missing outputs separate these means from the earlier draft; the change cannot be assigned to one lost beat alone.

A.2 Coverage variation and the historical zero-exclusion control

The generator excludes $\mathrm{round}(EH)$ of the H distinct required information holders from the bounded roster. Thus E controls holder exclusion; it is not an exact fraction of facts or information. Table 4 retains the reported coverage comparisons but separates the historical $E = 0$ experiment from the later nonzero settings.

For the nonzero rows, the available record does not identify the exact source batch, cell mapping or marginalization, and delivery filter. They cannot be treated as strata of the main factorial: its separately reported formation contrasts are 0.112 and 0.352 at low and high exclusion (Appendix F).

Table 4: Reported coverage comparisons on requirement F_1. The $E = 0$ row uses the historical three-arm runtime; it is not a missing cell of the current 2×2. The labels “open” and “bounded” describe reach in that historical comparison.

E	Open	Bounded	Difference	Bounded sources	95% CI
0 (historical)	0.823	0.789	+0.034	4.36	$[-0.067, 0.137]$
0.3	0.791	0.679	+0.111	3.17	$[0.020, 0.200]$
0.7	0.803	0.524	+0.279	1.26	$[0.210, 0.345]$

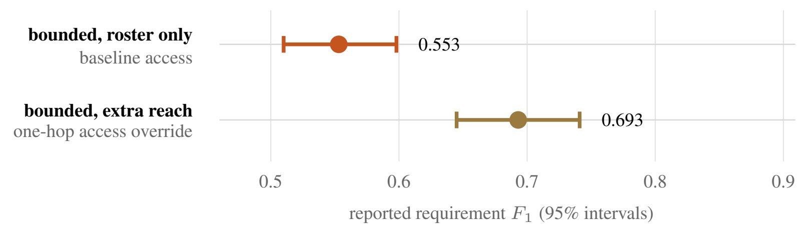

Figure 4: Additional reach improves the reported bounded-arm score in the relay sweep. Points and intervals are retained from the original figure; the comparison schedules 28 episodes per arm. Conditional on delivery, the reported difference is $+0.140$ (95% CI $[0.073, 0.204]$); scoring failed deliveries as zero gives $+0.088$ ($[0.016, 0.158]$). The implementation grants extra reach through an override, rather than realizing a complete broker, assent or responsibility-transfer process. Historical read-all and open configurations are not included in this paired comparison (Appendix A.3).

The high-exclusion discrepancy exceeds rounding. We retain both source summaries for traceability, without equating their estimands or claiming a reconciled coverage effect.

The smaller historical $E = 0$ contrast is consistent with a role for coverage, but its interval permits differences in either direction and does not establish equivalence. The historical runtime preceded the present Store list/read tools and strict Sandbox interface. Its “public” and “private” labels must not be retrospectively identified with current cells A and D. The 27-episode first-model control comprises three arms, three scenarios and three seeds, giving nine episodes per arm. The 18-episode second-model control is a separate comparison, not a replication of the current factorial effects.

Historical stage-specific token comparisons are discussed separately in Appendix F.2; they do not establish a T1 accuracy–cost ranking for the current A/D configurations.

A.3 An extra access path

In the reported `scaling` comparison, bounded Sandbox configuration C was evaluated with relay depth zero or one, balanced across three edge-cost settings, at 28 episodes per setting of relay depth. The delivered-output requirement F_1 was 0.553 without relay and 0.693 with relay; useful-source counts were 1.50 and 3.40. The reported difference was $+0.140$ (95% CI $[0.073, 0.204]$) conditional on delivery and $+0.088$ ($[0.016, 0.158]$) when missing outputs were scored as zero. These answer different questions and are retained together.

An earlier 16-per-arm comparison reported $+0.042$ ($[-0.137, 0.208]$); it is superseded in the study narrative, not a second estimate from the same denominator. The code permits an additional outside-roster query through a runtime override. It does not establish a real social contact graph or enforce a broker’s independent grant on that downstream query (Appendix H.4). The result therefore concerns extra access, not the full mechanism of a consent-based introduction.

The early read-all diagnostic listed 0.836 versus an open baseline of 0.857, while its text described a different contrast of $+0.019$. These values lack a common documented matching set. We do not combine them with the relay comparison or infer equality from them.

A.4 Friction, result caps and synthetic reputation

Per-edge friction. The study record reports open-minus-bounded differences of $+0.219$, $+0.263$ and $+0.244$ at zero, one and two handshake turns per outside-roster contact. The nonzero settings triggered 194 and 384 handshakes; reported token use rose from 53k to 74k. Within these tested settings, added friction increased expenditure without reversing the point-estimate ranking. This does not bound larger costs, hard budgets or the cost of maintaining a relationship.

Search cap. Caps of 5, 20 and 80 returned cards produced reported requirement F_1 values of 0.791, 0.755 and 0.819; the reported contrast intervals included zero. A cap is a limit on the cards returned by a search, not the total directory size and not a guarantee that every returned card is read. These estimates do not establish that search width is irrelevant.

Reputation marks. The off, real-mark and random-mark conditions yielded reported means of 0.821, 0.817 and 0.833, with contrast intervals including zero. "Real" here means a synthetic mark constructed from whether the card currently contains planted misinformation; it does not mean a reputation learned from repeated outcomes. Required facts generally have one designated holder, and a flagged card can also be that fact's only source. Avoiding it can therefore sacrifice necessary information. This limits what the null says about reputation-based choice among substitutable sources.

A.5 Withdrawn misinformation measurements

Distractors were part of the task construction, but the earlier absorption and fabrication counts are not retained as validated outcomes. Inspection reproduced two distinct measurement failures. Store previews can reveal a planted statement without logging it as seen, whereas asking a card logs all of its planted statements even if the reply omits them. Separately, matching numbers anywhere in raw HTML can count a CSS value or an unrelated number as an asserted task fact. Exposure state also resets between beats, despite persistent context in some configurations. These issues invalidate the interpretation of the previously reported 58-versus-1 fabrication comparison and its associated absorption rates. Full replies, artifact-level fact matching and history-aware exposure reconstruction are required before those claims can be used as evidence.

A.6 The withdrawn directory-size conclusion

The original `nscan` interpretation was that the open advantage disappeared as N increased from 50 to 800. That interpretation was withdrawn after missing artifacts had been scored as zero: the study record attributes 27% of beats to transport failures, compared with 2–3% in other sweeps. It also reports that scenario and execution time were confounded and that the two arms traversed the size settings in different orders. The retained delivery plot documents this problem rather than demonstrating a causal response to directory size.

The outage diagnosis does not establish that N is absent from the bounded arm's causal path. Although the roster contains 20 cards at every size, the generator uses an N-dependent random stream to sample its filler membership. Changing N can therefore change bounded search results. Removing failed outputs also need not preserve the original scenario or difficulty distribution.

Among delivered open outputs, the reported means across the five sizes were 0.800, 0.777, 0.758, 0.779 and 0.826. In one fully delivered subset with only three episodes per size, search precision fell from 0.528 to 0.386, approximately 27%, while reported tokens doubled. This subset is exploratory: it does not establish size invariance, and it cannot support a million-agent extrapolation. Delivery-rate warnings in the current analysis code are diagnostics, not proof that a retained comparison is unbiased.

A.7 Retrieval sensitivity model

The retained figure is a toy sensitivity calculation, separate from the experiments. Let m be the number of cards matching a query, h the number holding the needed information and k the number considered. Define $\rho \in [0, 1]$ as an interpolation parameter between a mean rank $(m + 1)/2$ and $(h + 1)/2$. This is not an empirically measured correlation coefficient. Mean rank alone does not determine a top-k probability. The calculation additionally assumes a uniform rank distribution with

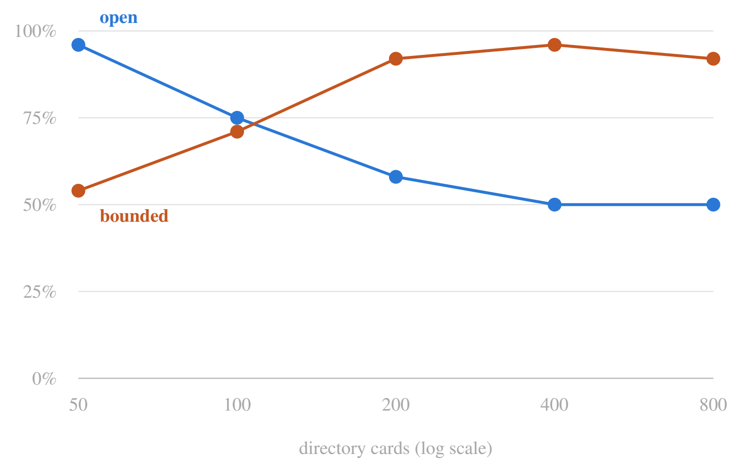

Figure 5: Reported delivery rates in the withdrawn directory-size sweep. The open and bounded configurations were evaluated in different time orders during a reported transport outage. These curves are retained as a diagnostic of missing outputs, not as evidence of a network scaling law. The original plot does not resolve the episode-versus-beat denominator; that denominator requires the run manifest. A fixed roster size does not hold its membership fixed when the generator changes N.

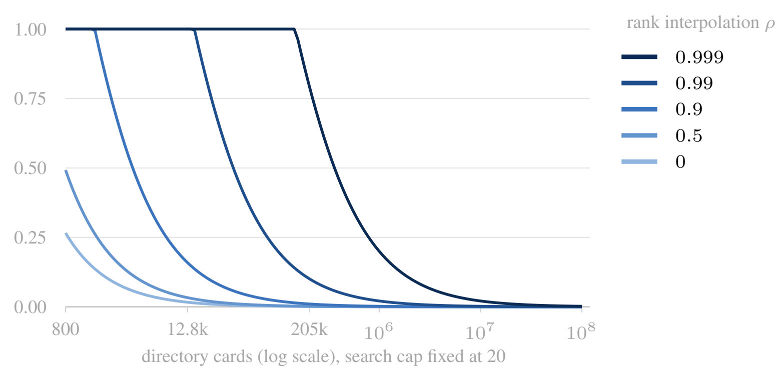

Figure 6: An illustrative retrieval model separates directory size from ranking assumptions. The vertical axis is modelled holder-retention probability at a fixed search cap of 20; the horizontal axis is directory size. Curves retain the original toy-model coordinates. Here ρ is a rank-interpolation parameter, not an empirically measured correlation or a fitted reputation score (Appendix A.7). Values near one delay the modelled loss of useful holders as the directory grows. The curves are illustrative, not fitted to experiment; their million-card behavior is not an empirical scaling law.

effective width $M = (1-\rho)m + \rho h$, giving the approximation

$$P(\text{rank} \le k) = \min\left\{1, \frac{k}{(1-\rho)m + \rho h}\right\}. \tag{3}$$

The original plotted coordinates correspond to $m = 0.094N$, $h = 6$ and $k = 20$; the repository's retrieval script uses a different holder count, $h = 2$. At $N = 10^6$ and $\rho = 0.999$, the plotted model gives a probability of approximately 0.20, not near-perfect retrieval. Its role is to illustrate dependence on ranking assumptions. It supplies neither an empirical ranking threshold nor evidence that feedback-based reputation solves large-scale retrieval.

B Implemented configurations and their boundaries

The four cells cross directory scope with the access interface. A and B can search the full directory, normally 100 cards; C and D search a roster of 20. A and C use Sandbox question answering. B and

D additionally expose Store enumeration and reading. A and D are the canonical public and private combinations used in this paper; B and C diagnose which parts of their difference depend on the implemented configuration dimensions.

Table 5: Configuration fields in the current runtime. A configured limit is not evidence that every corresponding mechanism was exercised successfully in the historical runs.

Field	**A/B: open**	**C/D: bounded**
Directory scope	All cards	Roster of 20
Namespace	Per contact	Persistent within episode
Responder history	Disabled	Enabled
Requester notes	Enabled	Enabled
Grant uses	One per issued grant	No configured use limit
Configured delegation depth	0	5
Access interface	A: Sandbox; B: Store	C: Sandbox; D: Store

Directory scope is bundled with namespace continuity and responder history. History can matter even within T1 when the same counterpart is contacted twice. The formation contrast is therefore an effect of this configuration bundle, not an isolated estimate of reach alone. The `sharedWorkspace` field is declared in the configuration; actual component persistence is controlled by namespace. Configured delegation depth also does not guarantee successful cross-specialist delegation under the current grant implementation.

Sandbox responders receive their local knowledge and answer the requested question. Store exposes the same card's short knowledge entries through `list_store` and `read_store`; list previews contain the first 60 characters. This is substantially less than unrestricted access to an enterprise folder. The Store interface also changes how builder outputs are returned and stored, so its effect includes the implemented handoff protocol. It should not be interpreted as a pure comparison between reading raw documents and asking questions about them.

The kernel mediates ordinary calls, but several boundary paths require correction before they can validate stronger enforcement claims. An out-of-roster Store request can raise an undefined-variable error instead of a normal denial. The relay path invokes an additional reachable target through an override rather than an independently enforced broker grant. These code-level findings do not reveal their incidence in the historical runs; that incidence requires the event logs. They do limit the mechanisms that the reported comparisons can be said to test.

C What the attribute intervention measures

The `seedaxis` experiment varies asserted relationship context while retaining the selected network configuration. Its six profiles are control, trust, accountability, origin, stranger and placebo. The implemented texts appear in Appendix H.3. They assert past accuracy, attribution or the origin of a relationship; they do not instantiate earned trust, incentives or a penalty for losing a relationship.

Origin and stranger apply opposite encounter narratives uniformly across the directory. Their comparison asks whether the framing itself changes behaviour. The placebo has the same number and placement of requester and responder lines as origin and stranger; it is not an exact token match for every profile. Trust uses three requester lines and accountability four responder lines. Shared goals and stated exit costs are possible attributes; enforcing an access penalty would also change formation or semantics. These were not separate conditions in this six-profile experiment.

This operationalization bounds the null result. Uniform descriptions provide no card-specific ordering, although they could still affect overall verification effort. The reputation sweep adds differential marks, but constructs them from planted content rather than accumulated outcomes. Neither intervention can establish that relational attributes in general are useless, or that stronger, actionable metadata would behave like these prompt assertions.

D Tasks, allocation and measured outcomes

D.1 One task family with three surface scenarios

All three scenarios require an agent to assemble distributed information into an HTML artifact with checked values, structure and calculations. The conference scenario builds a conference website with registration and related constraints; the lab-dashboard scenario combines experiment records, thresholds and metrics; the trip-planner scenario combines itinerary and budget information. Each defines two task instances, 14 facts per instance and four planted distractors. These are three topics within an information-pooling family, rather than independent evaluations of enterprise collaboration, personal errands and social interaction.

The standard timeline contains T1 (build instance A), T2 (revise A), T3 (build instance B) and T4 (revise B). All configurations retain requester notes. Bounded configurations can also retain responder history and components. T3 consequently tests reuse within an episode together with a new instance, not long-term relationship formation across tasks and users. The generator seeds on task instance as well as seed: roster membership can change from T1 to T3. A warm-stage difference therefore combines prior context with instance and allocation differences.

D.2 Directory construction and exclusion

The directory contains information holders, builders, near-miss cards and filler cards. The usual full directory has 100 cards and the bounded roster 20. Three random seeds control synthetic generation and holder allocation; they are independent of the textual relationship profiles. For H distinct required holders, $\mathrm{round}(EH)$ are excluded from the roster. At $E = 0$, all designated required holders are included, although retrieval and synthesis can still fail. Because holders carry unequal numbers of facts, $E = 0.3$ does not guarantee that exactly 30% of the facts are outside.

The bare and realistic profiles differ in distractor placement and relational metadata. The realistic profile redistributes planted falsehoods with respect to the roster when $E > 0$. It does not always place a falsehood on a different holder from its corresponding true fact. Thus changing E can also change the misinformation exposure distribution; the exclusion pattern alone does not identify a pure misinformation mechanism.

The eight-episode `k-pilot` varied the number of deliverable components. It was an exploratory test of composition and handoff overhead, not a test of access to complete folders or controlled cross-document reasoning. No new complex-task experiment is included in the reported episode totals.

D.3 Quality, cost and exposure

The reported quality measure is the implementation's *requirement* F_1. Its recall is the fraction of checks passed. Its precision-like term divides passed checks by checks classified as asserted, including certain wrong computed outputs. The harmonic mean is then reported as F_1. Checks combine HTML structure, text presence and numerical function outputs. This measure is not a general factual precision-and-recall score: errors outside the assertion set need not enter its denominator, and text matches can occur outside the intended field. Appendix H.2 specifies these limitations.

Recorded costs include input and output tokens reported for requester, responder and builder model calls, together with tool-call and contact counts. T1 cost and full-episode cost are distinct quantities. A crash path can replace the current beat's accumulated statistics with zeros; complete accounting therefore requires the event-level usage record as well as the summary. The historical efficiency table has no recovered common aggregation recipe, as detailed in Appendix F.

Distinct contacted counterparts and accessed resources describe the breadth of interaction and access. They are exposure proxies, not measurements of how much task information each party learned. A counterpart may see only a subquestion; a Store read may expose material to the requester without conveying the full goal to the resource owner. These directions of exposure should not be collapsed into a single privacy-loss claim.

The original draft also described realization, boundary cost and regime-crossing metrics as established outputs. The current implementation does not support those descriptions as validated measures. Blind scoring of utility, novelty and diversity was not run, and no empirical conclusion here relies on it.

E Comparisons and statistical interpretation

E.1 Questions and their current evidential status

Table 6 preserves the study's hypothesis labels to make changes in interpretation traceable. No recovered timestamped protocol establishes that all of these hypotheses, outcomes or analyses were fixed before the relevant runs. In particular, H3′ is a weaker, post hoc reformulation of H3 using a different outcome.

Table 6: Questions represented in the existing study. Status refers to the available reported results and measurement boundaries, not to a verified preregistration.

	Question	Current interpretation
H1	Is the formation contrast larger than tested attribute contrasts?	Larger reported T1 point estimate; scoped to this task and these interventions.
H2	Do relationship assertions produce stable gains?	No consistent gain in the reported profiles; not evidence of zero attribute effect.
H3	Does Sandbox outperform Store on quality?	Contradicted by the reported four-cell quality means.
H3′	Is Sandbox more efficient per token?	Post hoc descriptive comparison; cost window and aggregation require verification.
H4	Does coverage help explain the formation contrast?	Supported as an interpretation by exclusion and extra-access comparisons, with generation and implementation limits.
H5	Does Store increase misinformation adoption?	Withdrawn as an empirical claim because exposure and outcome measurements are invalid.

E.2 What the four cells identify

Let A, B, C and D denote cell means under the mapping in Appendix B. The reported configuration contrasts are

$$\Delta_F = \tfrac{1}{2}[(A - C) + (B - D)], \tag{4}$$

$$\Delta_S = \tfrac{1}{2}[(B - A) + (D - C)]. \tag{5}$$

The canonical public-minus-private comparison is $A - D = \Delta_F - \Delta_S$. The diagnostic diagonal is $B - C = \Delta_F + \Delta_S$. Both identities hold for any four means; neither tests additivity or the absence of interaction. The relevant interaction contrast is $(B - A) - (D - C)$. Its interval cannot be reconstructed from rounded cell means.

The factorial holds the model family, requester runtime, kernel, task family and scoring implementation in common. Configuration fields and access-specific instructions differ as described in Appendix B. Shared loop limits do not imply identical realized tokens or identical effective tool budgets: a requester iteration can contain multiple calls, and the composition setting adds iterations. The attribute sweep changes text within a fixed configuration rather than changing access permissions.

E.3 Repeated measurements and missing outputs

An episode, not a beat, is the unit that links T1–T4. A valid analysis of their joint trajectory must preserve this dependence. The current scripts use different input paths and resampling methods; the main factorial script does not reproduce the manuscript's stated two-batch delivered-only analysis. Counts of nominally significant contrasts are not a correction for multiple testing, and an interval including zero is not an equivalence test.

Conditioning on delivery estimates quality among retained outputs. Scoring missing outputs as zero estimates a different end-to-end quantity. Neither treatment alone identifies the effect of a configuration when missingness depends on task, timing or implementation failure. Transport failures, absent submissions, scorer errors and runtime crashes need separate accounting. An aggregate delivery-rate threshold cannot certify comparability, even if loss rates happen to match.

F Extended reported results

The main factorial values below follow the aggregation description in Appendix A.1; the attribute results come from the separate 72-episode `seedaxis` sweep. All retained empirical numbers are reported study estimates, not a new computation from unavailable raw runs.

F.1 Factorial means and diagnostic contrasts

Table 7: Reported T1 requirement F_1 and configuration contrasts. The study record assigns these means to the two pooled factorial batches, with 35–36 delivered outputs per cell. The precise denominators and interval-generating pipeline remain to be verified from the original runs.

	Sandbox	**Store**
Open	A: 0.808	B: 0.911
Bounded	C: 0.596	D: 0.627
$\Delta_F = 0.248$, reported 95% CI $[0.194, 0.302]$		
$\Delta_S = 0.066$, reported 95% CI $[0.012, 0.119]$		
Canonical $A - D = 0.181$; diagnostic $B - C = 0.314$		

The displayed means imply Store-minus-Sandbox differences of 0.103 within open and 0.031 within bounded configurations. Their difference is 0.072. This describes heterogeneity in the point estimates; without raw paired observations, it supplies no confidence interval or significance claim for the interaction. Small discrepancies between arithmetic on the displayed means and reported aggregate contrasts can arise from rounding and are not corrected by inventing additional precision.

The original summaries also report a larger formation contrast at high than at low exclusion (0.352 versus 0.112 at T1) and semantics contrasts of 0.031 on T2, 0.084 on T3 and 0.055 on T4. These are stage-specific estimates, not additional independent replications. The T3 comparison changes task instance as well as available history, and a known outdated trip-planner revision check affects interpretation of rework scores. The patterns motivate a task-boundary question but do not establish that one configuration wins at every stage.

F.2 Cost summaries with unresolved aggregation

The previous appendix reported formation and semantics token contrasts of -17.1k and $+11.0$k, described there as "per episode". The available factorial script instead reads T1 tokens. A later manuscript revision attributes an approximately 19k bounded–open increment to semantics, but supplies no reproducible calculation. That contrasts formation configurations, not Store with Sandbox, so it cannot replace the 11k estimate. The earlier appendix also reported the following efficiency values, for which a common calculation and filtering script has not been recovered:

	Sandbox	**Store**
Open	0.177	0.157
Bounded	0.094	0.087

These are historical values labelled F_1 per 10k tokens, retained for traceability. A ratio of mean quality to mean tokens and a mean of per-episode ratios are different estimands. Together with the unresolved time window and delivery filtering, this prevents using the table as a verified cost-efficiency ranking. Nor can an open-minus-bounded cost contrast be substituted for a Store-minus-Sandbox effect. The present manuscript therefore does not infer equal-reach public dominance or a universal cost for exposing state.

The historical Flash $E = 0$ record (`FINDINGS.md` at source revision `a7fdcf1`) reports T3 token means of 49.8k for public and 81.5k for private: a same-stage difference of 31.7k, approximately 64% more for private. This is a historical three-arm comparison, not the current A/D comparison. It cannot be combined with the T1 quality difference of 0.034 to infer equal quality at lower cost. The historical GLM record gives relative warm speedup rather than absolute cross-arm token totals; those totals cannot be inferred from the reported ratios.

F.3 Relationship text

Table 8: Reported relationship-text contrasts on requirement F_1, from 72 episodes across six profiles ($n = 12$ per profile in the study record). The reported estimator was stratified by arm and scenario. Parenthetical entries describe the original interval summary; exact endpoints are not supplied in that table.

Contrast	**T1**	**T3**	**Reported interval summary**
Placebo − control	$+0.005$	-0.014	Includes zero
Trust − control	-0.010	$+0.017$	Includes zero
Accountability − control	-0.046	$+0.015$	Includes zero
Origin − placebo	-0.088	$+0.074$	Positive T3 interval excluded zero
Stranger − placebo	-0.116	$+0.042$	Negative T1 interval excluded zero
Origin − stranger	$+0.029$	$+0.032$	Includes zero

The reported profiles did not yield a consistent positive effect across stages. The placebo comparison does not rule out every effect of prompt length: its matching applies to origin and stranger, and an uncertain small estimate is not proof that added text is inert. Similarly, counting the contrasts whose intervals exclude zero does not implement multiple-testing correction. The defensible conclusion is no consistent positive score effect across these profiles and stages, with substantial uncertainty about the magnitude and direction of effects under alternative placement, stronger signals or actionable source choices.

G Boundaries of the interpretation

G.1 Coverage and access play different roles

Failing to reach required holders directly costs quality in these tasks. Exclusion and extra-access comparisons suggest that reach contributes to the formation contrast, but cannot isolate it: responder history and namespaces are bundled with formation, profile-dependent distractor placement changes with exclusion, and the equal-reach control uses another runtime generation.

Store changes what a requester can inspect after reaching a source. Its short entries do not model folders of records, drafts and revisions. The reported benefit neither bounds nor establishes the value of access to such structures. The complex-task experiment remains unrun, with its section reserved without results.

G.2 Friction does not yet model relationship consequences

The default adds no onboarding charge; the edge-cost sweep adds handshake turns. Neither implements access negotiation, strategic refusal, maintenance costs or relationship loss after misuse. These mechanisms motivate the public/private distinction. The estimates concern operational proxies, not the full economics of real networks [Jackson and Wolinsky, 1996].

The open configuration's advantage is no upper bound on deployment performance: omitted factors could favour either arrangement. Contact counts indicate interaction scope; they neither price confidentiality nor establish Pareto dominance across quality, cost and exposure.

G.3 A null for text is not a null for relationships

Uniform trust or history claims provide little differentiation for source selection. Synthetic marks address that limitation, but a unique source per fact cannot easily be replaced without losing information. These constraints qualify the negative results; earned reputation and enforceable accountability remain largely untested. The third axis therefore remains in the framework.

G.4 Scope of the evidence

The second-model $E = 0$ control tests the historical three-arm setup, not model robustness of the four-cell effects. Four beats permit revision and reuse, but establish no autonomous relationship formation, feedback accumulation or relationship loss across episodes. The withdrawn size sweep and retrieval model likewise cannot demonstrate behaviour at very large scale.

Deterministic scoring remains valid only when its assertions measure the intended outcomes. Reproduced text-matching, revision-key and exposure-tracking errors limit current claims. Repairing code cannot repair historical estimates without original artifacts and assertion versions; exposure reconstruction also requires full messages. These defects do not determine the direction of corrected effects.

H Source definitions and reproducibility details

H.1 Question-to-evidence map

Evidence locations: the factorial, Figure 2 and Table 7; coverage, Appendix A.2; extra access, Appendix A.3; relationship text, Table 8; reputation, Appendix A.4; withdrawn scaling, Appendix A.6. This is a retrospective map, not a recovered pre-run plan.

H.2 Scoring implementation

The scenario files specify the required HTML structure and numerical functions. The scorer evaluates DOM existence and count checks, text checks and calls to the requested function with reference outputs. Its text checks use substring matching over body text, which can include script content and does not generally bind a value to a particular field. Thus an unrelated number or a longer number can satisfy a check. The reported requirement F_1 in `src/lib/timeline.js` is

$$r = \frac{n_{\text{pass}}}{n_{\text{checks}}}, \qquad p = \frac{n_{\text{pass}}}{n_{\text{asserted}}}, \qquad F_1 = \frac{2pr}{p+r}, \tag{6}$$

with zero when a denominator or $p + r$ is zero. “Asserted” includes passed checks and mismatching checks whose returned value is not null. This differs from exhaustively identifying all correct and incorrect factual statements in an artifact.

Revision assertions are formed by replacing specified base checks and appending new checks; the current helper has no deletion operation. In the trip-planner revision, two checks for superseded museum information remain active. A correct revision can therefore lose those checks and be counted as a regression. Code inspection reproduced this failure and also verified 86 numerical calculation assertions against their reference implementations. That positive calculation check does not validate the text or revision scoring rules, nor does it determine corrected scores for unavailable artifacts.

The online and offline rescoring paths also differ in some assertion handling and in which summary fields they refresh. A reproducible revision of these results needs one scoring definition, versioned assertion sets, and regeneration of derived summaries from the retained artifacts. No claim in this revision assumes that this repair has already been completed.

H.3 Implemented relationship texts

These synthetic assertions are transcribed from `src/lib/seeds.js`, preserving grouping and placement. They are not verified histories. Control adds no text; requester and responder receive only their respective text.

Trust: requester only.

> RELATIONSHIP CONTEXT: across your last three tasks, the agents you reached through this directory gave you accurate values every time. You did not need to cross-check any of them against a second source.

Accountability: responder only.

> ACCOUNTABILITY: this answer is logged and attributed to you by name. Its accuracy is written back to a record other agents read when deciding whether to consult you. An answer you are not sure of costs you more than saying you do not hold it.

Origin: requester, then responder.

> RELATIONSHIP CONTEXT: the agents in this directory are colleagues you have worked alongside for two years.
>
> The agent contacting you is a colleague you have worked alongside for two years.

Stranger: requester, then responder.

> RELATIONSHIP CONTEXT: you found these agents on an open board. You have never worked with any of them.
>
> The agent contacting you is a stranger you have never worked with.

Placebo: requester, then responder.

> OPERATING CONTEXT: this directory was last reindexed on a Tuesday. Index maintenance has no bearing on what any agent holds.
>
> The directory you are listed in was last reindexed on a Tuesday.

H.4 Relay implementation

The current relay path offers outside-roster candidates from the global directory, rather than from a separately generated acquaintance graph for each broker. A broker can select an outside target, after which the runtime issues another contact with `forceReachable=true`. Although the code constructs a relay grant, that grant is not used to authorize the downstream call; the call is authorized directly for the requester. The downstream answer is returned with a relay prefix, rather than being paraphrased by another model call. Accordingly, the comparison measures an additional query path and its cost. It does not measure authentic introductions, consent propagation or cumulative paraphrase loss.

H.5 Code and data needed to reproduce the reported tables

The inspected source, revision `83513ded`, specifies Node.js ≥ 20.11, SharedOS `0.1.0-alpha.2` and default model `deepseek/deepseek-v4-flash`. These defaults do not establish every historical run's software or model version. Defaults are a 100-card directory, 20-card roster, search cap 40, no handshake turns or relay, and one deliverable component. The requester starts with 22 iterations and receives four more per added component.

Source locations are `src/lib/arms.js` for configurations, `src/lib/directory.js` for allocation, `src/scenarios/` for tasks, `src/lib/seeds.js` for relationship text, and `src/score.js` with `src/lib/timeline.js` for scoring. The runtime writes JSON summaries, JSONL events, HTML outputs and assertion snapshots, not a separate experiment database.

No verified command regenerates all retained values. Reproduction requires an episode manifest; batch commands, commits, models and profiles; per-beat status and token usage; artifacts and assertion snapshots; full messages for exposure analysis; and a script specifying each table's pooling, matching, delivery filtering and resampling. The run-level `all.json` can be overwritten and omit crashes, so it cannot replace a complete attempt ledger. Until reconciled, the tables remain reported results.